

Task 1: Role-Based Access Control (RBAC)

Step-by-Step Configuration:

1. In the Azure Portal, search for and select **Resource Groups**.
2. Select your existing Resource Group.
3. On the left-hand menu, click **Access control (IAM)**.
4. Click **+ Add > Add role assignment**.
5. **Assign Roles:** Repeat this for each user in the table:
 - **Role:** Search for the role such as the *Contributor*.
 - **Members:** Select **+ Select members** and find the specific user account
 - **Review + assign:** Complete the process.
6. **Verification:** Log out and log back in as user-audit (the Reader). Try to create a new resource (like a storage account) inside the Resource Group.

Microsoft Azure

Dashboard > Techbridge-RG | Access control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

Role

Contributor

Scope

/subscriptions/99a5f00b-d7d7-4b59-886f-83b5d137bc38/resourceGroups/Techbridge-RG

Members

Name	Object ID	Type
Alex Mokoena	57f104dd-8df2-465c-b4d1-3e2319ec4f3a	User

Description

No description

Review + assign

Previous

Next

Feedback

Microsoft Azure

Dashboard > Resource Manager | Resource groups > Techbridge-RG | Access control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

Role

Billing Reader

Scope

/subscriptions/99a5f00b-d7d7-4b59-886f-83b5d137bc38/resourceGroups/Techbridge-RG

Members

Name	Object ID	Type
Morgan Smith	2c959e6e-af3c-4487-8d8c-633303ea44b8	User

Description

No description

Review + assign

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Microsoft Azure

Dashboard > Resource Manager | Resource groups > Techbridge-RG | Access control (IAM)

Add role assignment

Role Members Conditions Review + assign

Role
Contributor

Scope
/subscriptions/99a5f00b-d7d7-4b59-886f-83b5d137bc38/resourceGroups/Techbridge-RG

Members

Name	Object ID	Type
Jordan Nkosi	f93504da-e488-406c-8863-be03d2a3dba3	User

Description
No description

[Review + assign](#) [Previous](#) [Next](#) [Feedback](#)

Microsoft Azure

Dashboard > Resource Manager | Resource groups > Techbridge-RG | Access control (IAM)

Add role assignment

Role Members Conditions Review + assign

Role
Reader

Scope
/subscriptions/99a5f00b-d7d7-4b59-886f-83b5d137bc38/resourceGroups/Techbridge-RG

Members

Name	Object ID	Type
Sam Taylor	93fc743e-ba2d-4d4b-b6bd-dca582354818	User

Description
No description

[Review + assign](#) [Previous](#) [Next](#) [Feedback](#)

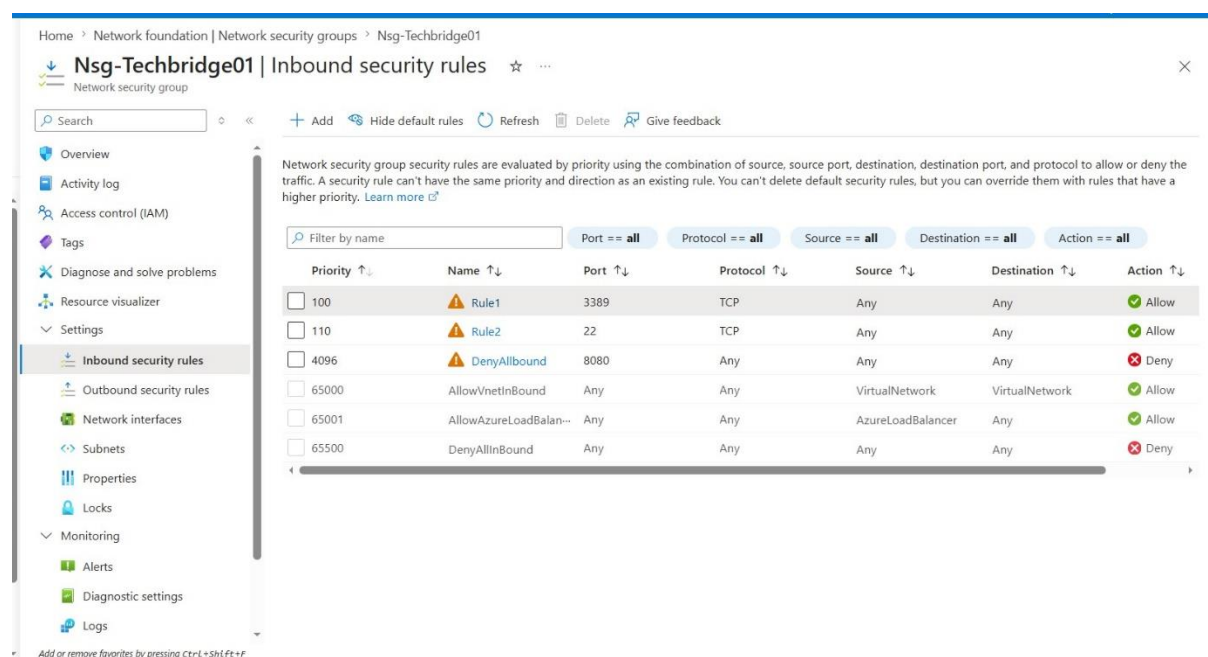
Assigning at the **Resource Group scope** ensures the user only has power over specific project files. If assigned at the **Subscription scope**, that user would have "Contributor"

rights over *every* resource group in the subscription, violating the principle of least privilege.

Task 2: Network Security Group (NSG)

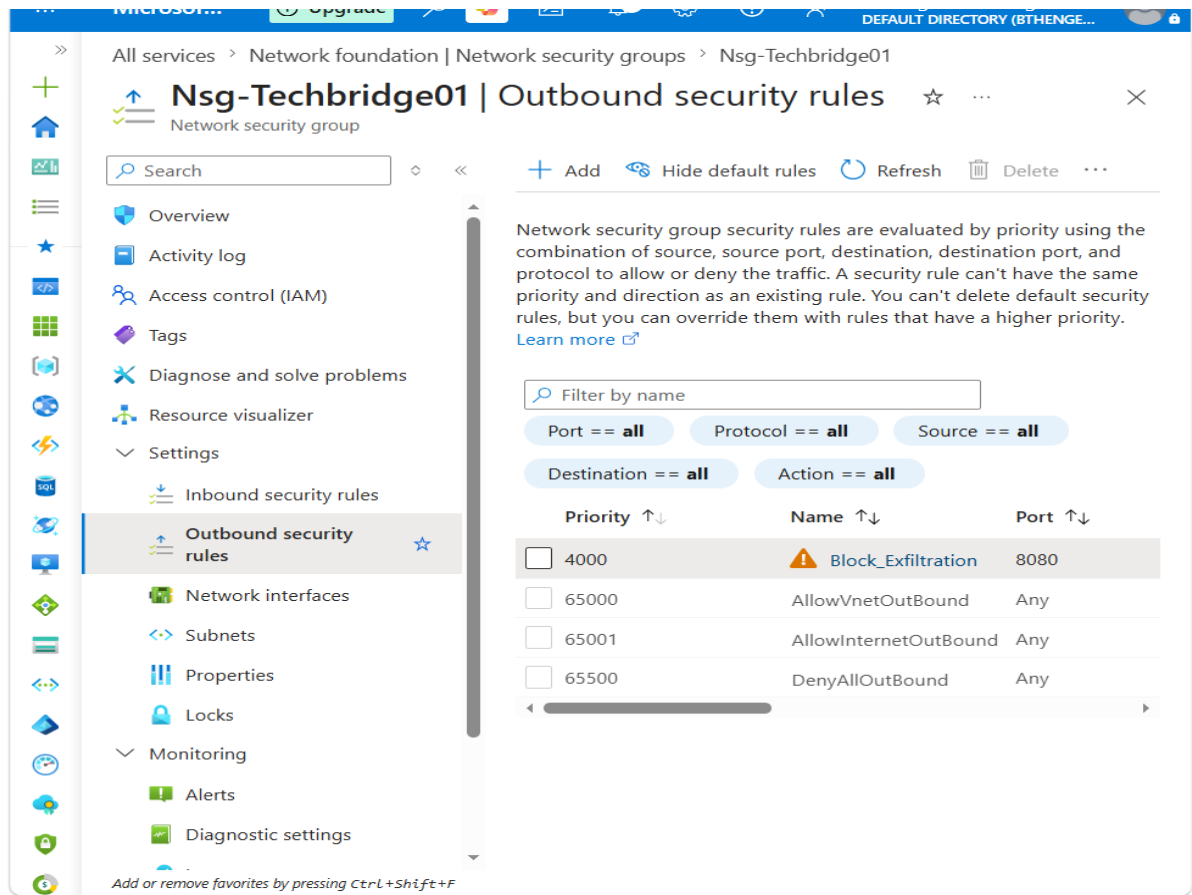
Step-by-Step Configuration:

1. Search for **Network security groups** and click **+ Create**.
2. **Basics:** Select your Resource Group. Name it NSG-Techbridge01. Click **Review + create**.
3. Go to the new NSG resource. Select **Inbound security rules** from the left menu.
4. **Add Inbound Rules:**
 - **Rule 1 (Management):** Port: 3389 (RDP) or 22 (SSH) / Action: Allow / Name: Rule 1
 - **Rule 2 (Web):** Port: 80, 443 (HTTP/S) / Action: Allow / Name: Rule 2.
 - **Rule 3 (Deny All):** Port: Any / Priority: 4096 / Action: Deny / Name: DenyAllInbound.



5. Add Outbound Rule:

- Destination: Service Tag / Dest Service Tag: Internet / Action: Deny /Name: Block_Exfiltration.



6. **Associate:** Click **Subnets** (under Settings) > **+ Associate**. Select you're the previous Virtual Network and Subnet.

Task 3: Storage Account & Containers

Step-by-Step Configuration:

1. Search for **Storage accounts** and click **+ Create**.
2. **Basics:**
 - o **Name:** Must be globally unique **Techbridge-Acc**
 - o **Redundancy:** Select **LRS (Locally-redundant storage)**.
3. **Review + create** and wait for deployment.

Microsoft Azure Upgrade

Home > Storage center | Blob Storage

Create a storage account

Subscription * Azure subscription 1

Resource group * Techbridge-RG
[Create new](#)

Instance details

Storage account name * techbridge01

Region * (US) Central US
[Deploy to an Azure Extended Zone](#)

Preferred storage type Choose preferred storage type

Performance *

☒ **Standard:** Recommended for most scenarios (general-purpose v2 account)

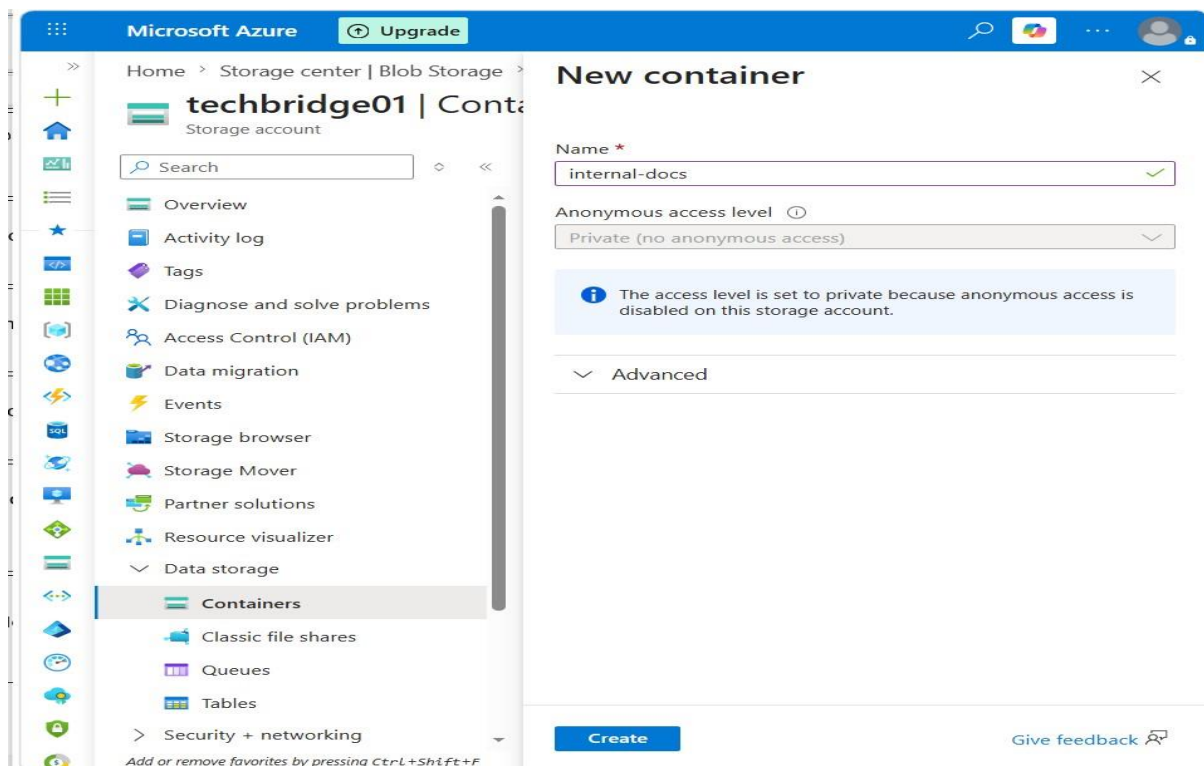
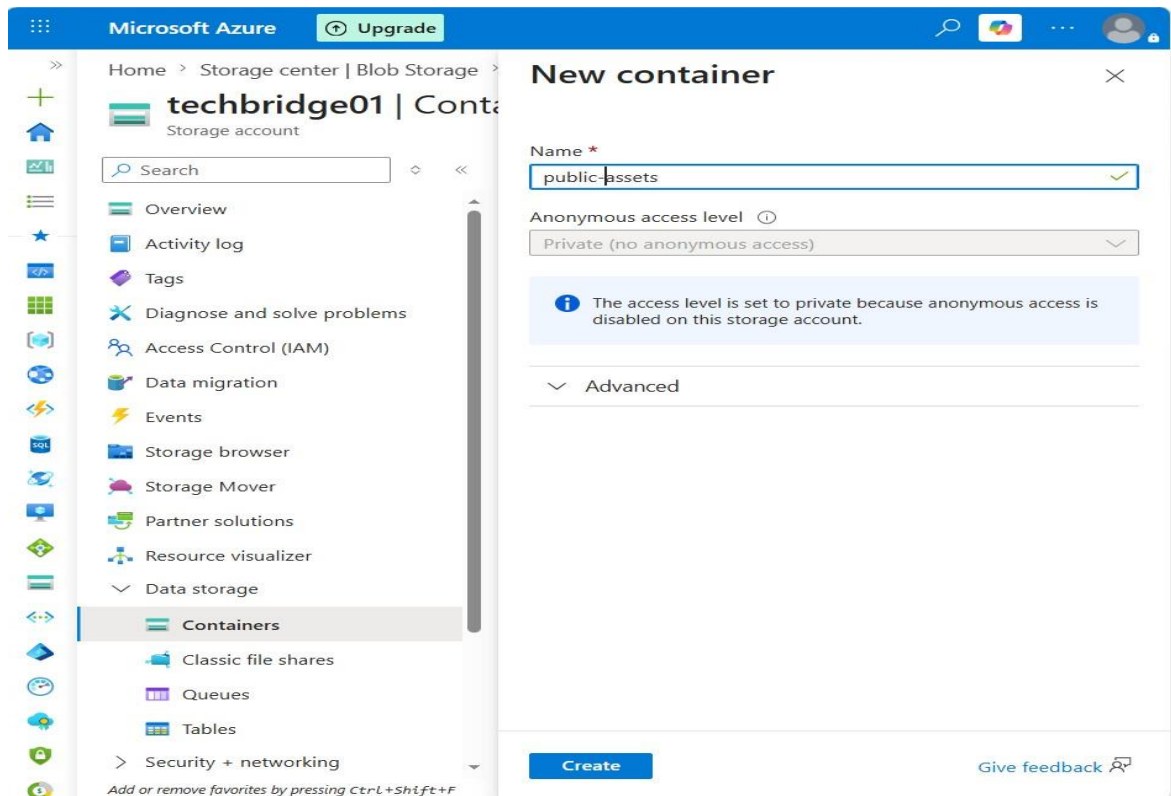
☐ **Premium:** Recommended for scenarios that require low latency.

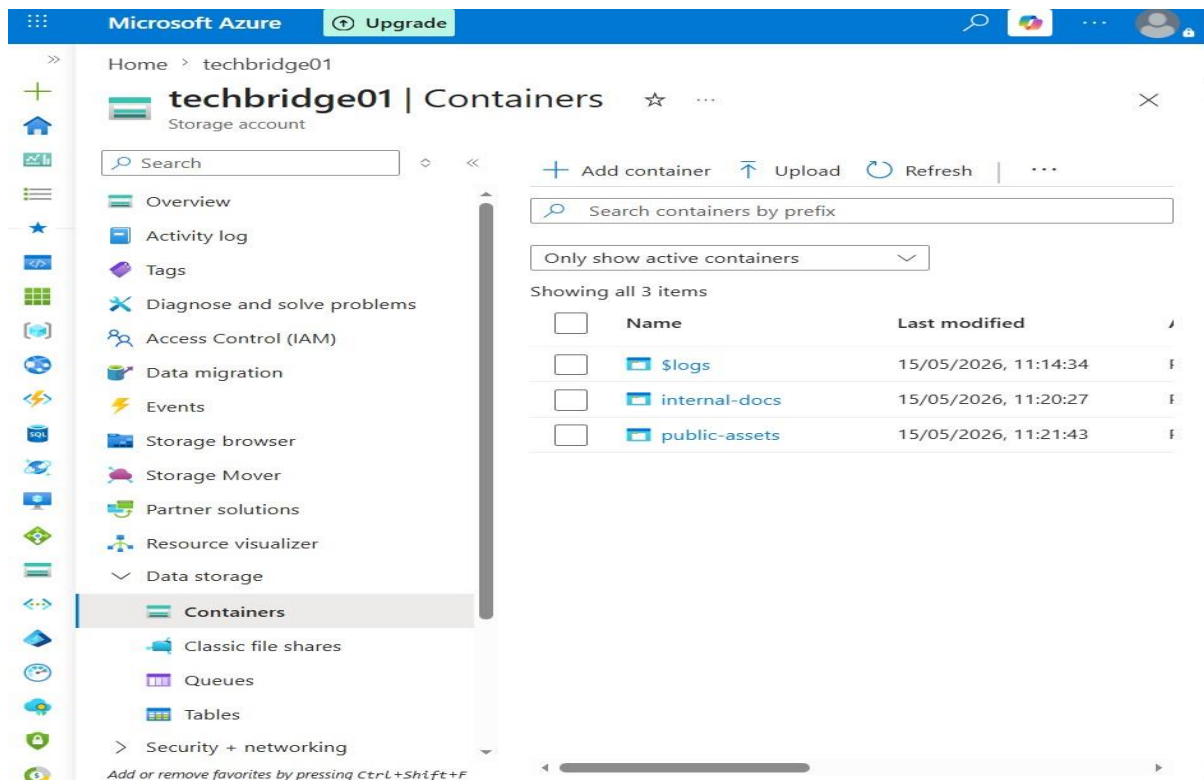
Redundancy * Locally-redundant storage (LRS)

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

4. Go to the Storage Account > **Data storage** > **Containers**.
5. Click **+ Container**.
 - Create internal-docs (Public access level: **Private**).
 - Create public-assets (Public access level: **Private**).

6. Click into each container and select **Upload**





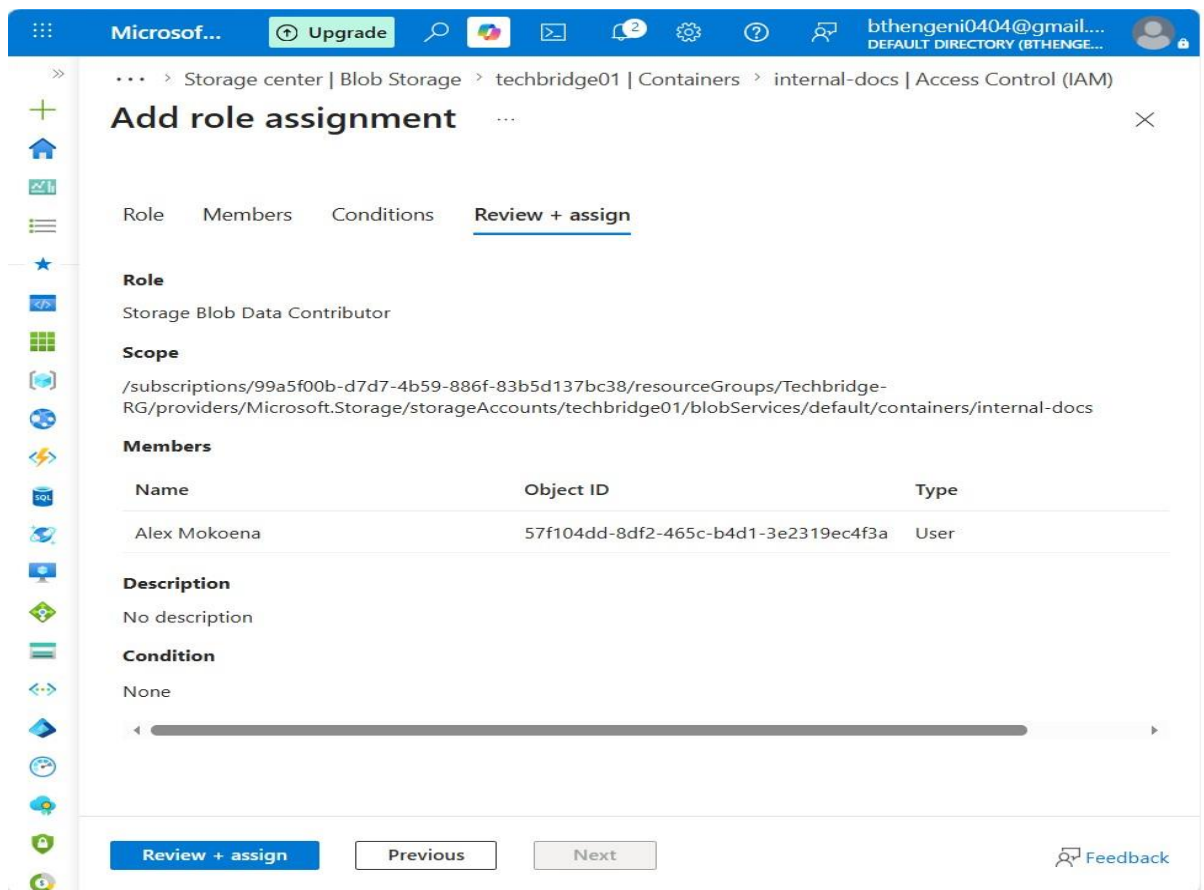
LRS is the most cost-effective option and is appropriate here because the scenario specifies a "small organization" with "no disaster recovery requirement."

Task 4: Storage Access Permissions

Step-by-Step Configuration:

1. RBAC Access (Container 1):

- Go to your internal-docs container.
- Select **Access Control (IAM)** > **+ Add role assignment**.
- Assign Storage Blob Data Contributor to A user.



2. SAS Token (Container 2):

- Go to your public-assets container.
- On the left menu, select **Shared access tokens**.
- **Permissions:** Select Read.
- **Expiry:** Set it for **24 hours** from now.
- Click **Generate SAS token and URL**. Copy the **Blob SAS URL**.

Microsoft Azure

Upgrade

Home > techbridge01 | Containers > public-assets

public-assets | Shared access tokens

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Access policy

Properties

Metadata

Signing method

☒ Account key ☐ User delegation key

Signing key ⓘ

Key 1

Stored access policy

None

Permissions * ⓘ

Read

Start and expiry date/time ⓘ

Start

05/15/2026 11:25:53 AM

(UTC+02:00) Harare, Pretoria

Expiry

05/15/2026 7:40:53 PM

(UTC+02:00) Harare, Pretoria

Allowed IP addresses ⓘ

for example, 168.1.5.65 or 168.1.5.65-168.1....

Allowed protocols ⓘ

☒ HTTPS only ☐ HTTPS and HTTP

Generate SAS token and URL

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